

Similarly, we can show that u is not a scalar multiple of v .

$\therefore \{u, v\}$ is linearly ~~linear~~ independent.

$\{u, w\}$: Let w be a scalar multiple of u . $\therefore \exists c \in \mathbb{R}$ s.t. $w = cu$

$$\begin{bmatrix} 0 \\ -5 \\ 2 \end{bmatrix} = c \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 \\ -5 \\ 2 \end{bmatrix} = \begin{bmatrix} 3c \\ 2c \\ -4c \end{bmatrix} \quad (\text{Scalar multiplication})$$

$$3c = 0 \Rightarrow c = 0 \quad 2c = -5 \Rightarrow c = -5/2. \text{ This is a contradiction}$$

$\therefore w$ is not a scalar multiple of u . Similarly, we can show that u is not a scalar multiple of w .

$\therefore \{u, w\}$ is linearly independent.

$\{u, z\}$: Let z be a scalar multiple of u . $\therefore \exists c \in \mathbb{R}$ s.t. $z = cu$

$$\begin{bmatrix} 3 \\ 7 \\ -5 \end{bmatrix} = c \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 \\ 7 \\ -5 \end{bmatrix} = \begin{bmatrix} 3c \\ 2c \\ -4c \end{bmatrix} \quad (\text{Scalar multiplication})$$

$$\therefore 3c = 3 \Rightarrow c = 1 \quad 2c = 7 \Rightarrow c = 7/2. \text{ This is a contradiction}$$

$\therefore z$ is not a scalar multiple of u . Similarly, we can show that u is not a scalar multiple of z .

$\therefore \{u, z\}$ is linearly independent

Note: It's getting a bit repetitive. We can similarly prove that $\{v, w\}$, $\{v, z\}$ and $\{w, z\}$ are each linearly independent. $\square$

b) Does the answer to part (a) imply that $\{u, v, w, z\}$ is linearly independent?

Solution: If $c_1u + c_2v + c_3w + c_4z = 0$ if and only if $c_1 = c_2 = c_3 = c_4 = 0$, then $\{u, v, w, z\}$ is linearly independent. $\therefore$ We cannot conclude anything from the answer of part (a) regarding the linear independence of $\{u, v, w, z\}$.

c) To determine if $\{u, v, w, z\}$ is linearly dependent, is it wise to check if, say, w is a linear combination of u, v and z ?

Solution: According to Theorem 7, $\{u, v, w, z\}$ is linearly dependent if and only if **some** vector is a linear combination of the